Indian Institute of Technology (ISM) Dhanbad
Department of Mathematics and Computing

MCC505 – Probability and Statistics

Monsoon Semester 2024-25

Mid Semester Exam | Total Marks: 32 | Date: 18/09/2024 (10 AM to 12 Noon)

Instructions:

(i) Solve all questions. All the parts of each question should be solved in one place.
(ii) Write your roll number, batch, group, course code, course name, etc. on your answer sheet.
(iii) Calculator is allowed. Mobile/Smartwatches/any other electronic item, etc., are not permitted in the exam hall.
(iv) Mention the results/statements/theorems used to solve the problems for step marking.

1.(a) For a frequency distribution given below, calculate Mode, Q_3, D_7 and P_{49}.

Monthly Sales (Rs. Lakh)	Less than 20	Less than 30	Less than 40	Less than 50	Less than 60	Less than 70	Less than 80	Less than 90	Less than 100
No. of Firms	30	225	465	580	634	644	650	665	680

(4 marks)

(b) For a set of 10 observations on a variable X, the following calculations were made:

$$V=\frac{(X-30)}{2},\quad \sum_{i=1}^{10} V_i=-1,\quad \sum_{i=1}^{10} V_i^2=49.$$

It was latter noticed that an observation 40 was misread as 30. Find the correct coefficient of variation of the variable X. 74 (2 marks)

(c) The first moment about origin of a distribution is 5. The second and third moments about the mean are 20 and 140 respectively. Find the third moment of the distribution about 10. $\frac{-5}{45}$ (2 marks)

2. (a) Two friends decided to meet each other between 9:30 PM and 11:00 PM with the condition that each one will wait for the other for 15 min. What is the probability that they will meet? (2 marks)

(b) On the morning of January 1, a hospital nursery has 3 boys and some number of girls. That night, a woman gives birth to a child, and the child is placed in the nursery. On January 2, a statistician conducts a survey and selects a child at random from the nursery (including the newborn and every child from January 1). The randomly selected child is a boy. What is the probability the child born on January 1 was a boy? (4 marks)

(c) Prove that for n events $A_1, A_2 \dots, A_n$: $P(\cap_{i=1}^{n} A_i) \geq \sum_{i=1}^{n} P(A_i) - (n-1)$. (2 marks)

[P.T.O.]

3. (a) If X and Y are independent random variables, X takes the values 2, 5, and 7 with probabilities $\frac{1}{2}, \frac{1}{4},$ and $\frac{1}{4}$ respectively and Y takes the values 3 and 5 with probabilities $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Find the joint distribution of X and Y. Find the mean and variance of X & Y. (2 marks)

(b) Suppose a two-dimensional continuous random variable has a probability density function

$$f(x,y) = \begin{cases} kx(x-y), & 0 < x < 2 \text{ and } -x < y < x \\ 0, & \text{elsewhere} \end{cases}$$

Evaluate the constant k. Find the marginal distribution of X and Y. (2+2+2=6 marks)

4. (a) Does there exist a random variable X with mean μ_x and standard deviation σ_x for which $P(\mu_x - 2\sigma_x \leq X \leq \mu_x + 2\sigma_x) = 0.6$? Justify (2 marks)

(b) Suppose that the number of cans of soda pop filled in a day at a bottling plant is a random variable with an expected value of 10,000 and a variance of 1000. Obtain an upper bound on the probability that the plant will fill more than 11,000 cans on a particular day. What can be said about the probability that the plant will fill between 9,000 and 11,000 cans on a particular day? (2+2=4 marks)

(c) Let X_i assumes the values i and $-i$ with equal probabilities for $i = 1,2,\ldots,n$. Check whether mean $(\bar{X}_n)$ of the sequence of independent random variables $\{X_i\}$ obeys the weak law of large numbers. (2 marks)

******* End *******

Indian Institute of Technology (Indian School of Mines) Dhanbad
Department of Computer Science and Engineering
Principles of Artificial Intelligence (CSO303)
Mid Semester Exam, Date: September 15, 2024

Timing: 01:30 PM to 03:30 PM   Monsoon Semester (2023-2024)   Maximum marks: 60

1. (a) Compare atomic, factored, and structured representations. Provide an example for each.
   (b) Agents operate in various types of environments. Explain the differences between a deterministic and a stochastic environment. How does the nature of the environment influence the design of an intelligent agent?
   (c) A financial trading agent operates in stock markets and makes decisions on buying and selling stocks autonomously. Define the PEAS description for this agent. What metrics would indicate that the trading agent is performing well (e.g., profit maximization, risk minimization)? What aspects of the financial environment does the agent need to consider (e.g., market fluctuations, economic news)? What actions can the agent perform to execute trades or manage portfolios? What data sources and sensors would the agent rely on to make decisions (e.g., stock prices, economic indicators)?

(3+3+4)

2. (a) What is a problem-solving agent, and how does it differ from other types of agents? Define the terms state space, initial state, and goal state in the context of the search problem.
   (b) Differentiate between well-defined and ill-defined problems. Provide an example of each.
   (c) How does abstraction help in solving complex problems? Provide an example where abstraction can be applied to simplify a real-world problem.

(5+3+2)

3. Suppose you are given the following search space:

State	Next	Cost
P	Q	5
P	R	2
Q	S	6
Q	T	7
R	U	3
R	V	4
S	G	4
T	G	3
U	W	5
V	G	7
W	G	6

   (a) Draw the state space for the search problem, with P as the initial state and G as the goal state. Clearly indicate all paths and their respective costs.
   (b) Assume the initial state is P and the goal state is G. Using the search space from part (a), show how each of the following search strategies would expand the search tree to find a solution:

- Breadth-first search (BFS)
- Depth-first search (DFS)
- Uniform-cost search (UCS)
- Iterative deepening search (IDS)

At each step, show which node is being expanded and maintain a fringe (frontier) table. Also, indicate the final solution found by each algorithm and the cost associated with the path. Compare the results of the algorithms based on time complexity, space complexity, and optimality in the context of this search space. Why do some algorithms find a solution more efficiently than others?

(2+8)

4. (a) Define the time complexity of BFS in terms of the branching factor ***b*** and the depth ***d*** of the shallowest solution. How does the branching factor impact the performance of BFS?

   (b) Discuss how Uniform Cost Search (UCS) ensures finding the optimal solution in weighted graphs, particularly focusing on the importance of non-negative edge costs. Additionally, explain how the behavior of UCS would change if the graph contained negative edge costs.

   (c) Why is random-restart more effective in overcoming the limitations of basic hill climbing as opposed to stochastic hill climbing? Under what conditions would random-restart hill climbing be less effective? Provide an example scenario.

(2+3+5)

5. (a) Explain what a heuristic function is in the context of informed search strategies. Discuss the characteristics that make a heuristic admissible and consistent.

   (b) For the following scenario, design two possible heuristics (h1 and h2) for a robot navigating a city grid where it can only move in four directions (up, down, left, right). State which of your heuristics is admissible and consistent, and which one dominates the other with reasons.

   (c) In heuristic-based search algorithms like A*, the choice of the heuristic function directly affects performance. Provide examples of a well-designed heuristic and a poorly designed heuristic for the traveling salesman problem (TSP). What are the specific characteristics of each heuristic that lead to good or poor performance?

(3+3+4)

6. (a) Compare and contrast A* search and Greedy Best-First Search in terms of their use of heuristic functions, efficiency, memory usage, and optimality.

   (b) Sketch a proof of the optimality of A* under an admissible heuristic tree-search paradigm. Why is consistency more critical in graph-based A* search compared to tree-based search?

(4+6)

भारतीय प्रौद्योगिकी संस्थान (भारतीय खनि विद्यापीठ), धनबाद

Indian Institute of Technology (Indian School of Mines), Dhanbad

DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

Mid-Semester Examination, Monsoon [2024-2025]

Date: 21-09-2024 **Time: 2 hours** **Full marks: 32**

Course Name: Sociology of Health **Code: HSO404**

All questions are compulsory

1. Describe, in detail, the history of heath and healing during Neolithic period and Industrial revolution. [5 + 5 = 10]

2. Discuss the Identity model and Limits model of disability. [5 + 5 = 10]

3. Describe five features/assumptions of medical model of healthcare. [5]

4. Write on the following:

   a. The story behind 'Rosie the Riveter' [4]

   b. The role of parental screening in preventing disability [3]

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Semester: Monsoon **Session: 2024-25**
Examination: MID SEMESTER EXAM, B. Tech (Mining Engineering) **Semester: V**
Subject: SURFACE MINING (MNC 300) **Time: 2 Hours** **Max. Marks: 45**
Instruction: Answer **ANY THREE** questions

Q. No.	Question	Marks
1.	c) What is 'box-cut'? What are its objectives? With the aid of suitable sketches, describe the different types of box cut and also indicate their respective applicability and limitations. d) With the aid of plan, section and LVS describe briefly the parameters of the box-cut.	10+5
2.	Concerning the ripper operation on a horizontal area derive the expressions for followings, with the help of suitable sketch, a) optimum spacing between parallel passes and b) optimum depth of effective loosening in terms of known parameters	15
3.	Estimate the number of drills required to achieve a production of 3 million ton/year of coal from a surface mine having coal seam of thickness 10 m. The coal seam is flat and horizontal and the stripping ratio of the mine is $3 m^3/ton$. (Assume relevant data wherever necessary and mention them)	15
4.	Estimate the HEMM required in a surface Iron ore mine for producing 8 million ton of ore per year. The average stripping ratio is 0.4 ton/ton. The mine will be worked by a shovel dumper combination with a bench height of 12 m. (Assume relevant data wherever necessary and mention them)	15

Indian Institute of Technology (Indian School of Mines) Dhanbad
Department of Mining Engineering
Midsem Examination Monsoon Semester 2024-2025
Course for 3rd Year (5th Semester) B.Tech. (Mining Engineering) Students (103 nos.)
Subject: Mine Planning and Economics (Code: MNC 301)

Time: 120 minutes (10:00AM–12:00PM) | **Venue:** NLHC LH 05-07 | **Date:** 22-09-2024 | **Maximum Marks:** 64

*Instruction: Please mention your **Serial No.** on the top of answer sheet as per Attendance List*
*Answer all questions of the **three** sections and **marks** are assigned against each section*
*Use of **Scientific Calculator** is allowed*

Section A: Numerical Answer Type Questions – 36 marks
*Answer all the questions and **marks** are assigned against each question*

1. **(a)** The real rate of return (r_r) from a mining project is 14%. If the inflation rate (r_i) over the entire life of the mine is 5.5 %, then estimate the nominal rate (r_n) of return (in %). Use: $(1+ r_r)*(1+ r_i)=1+ r_n$

(b) Data from two production faces of an open pit iron ore mine are given.

Item description	Face 1	Face 2
Maximum production capacity (tonne/day)	1600	2000
Fe (%)	63	58
Production cost of ores (in INR/tonne)	1500	1200

Ores from two different faces are blended and supplied with Fe grade not less than 60 %. Based on the demand, the combined production is limited to a maximum of 2500 tonne/day. If the selling price of blended iron ores is INR 4500 /tonne, the optimal production from two faces in tonne/day, for maximizing the profit.

(c) A coal mining company examines the option of buying two types of dumpers with the following details.

Parameter	Type-1	Type-2	Constraints
Capital cost per dumper (in ₹ crore)	3.0	4.0	Maximum capital available for purchasing is ₹120 crore
Capacity in tonne	40	50	Minimum daily tonnage to be hauled is 31,000
Daily trips for each dumper	20	20	
Operating cost (in ₹) per tonne	300.0	200.0	

What are the optimum fleet of dumpers of Type-1 and Type-2 in order to minimize the operating cost?

(d) Assume that the interest rate is 5% compounded daily. What would be the equivalent simple interest rate?

(1 + 2 + 2 + 2 = 7 marks)

2. A Rs. 100 investment cost has been incurred at time t = 0 as part of a project having a 5-year lifetime. The salvage value is zero (assuming straight line depreciation). Project income is estimated to be Rs. 80 in year 1, Rs. 84 in year 2, Rs. 88 in year 3, Rs. 92 in year 4, and Rs. 96 in year 5. Operating expenses are estimated to be Rs. 30 in year 1, Rs. 32 in year 2, Rs. 34 in year 3, Rs. 36 in year 4, and Rs. 38 in year 5. The effective income tax rate is 32%. Calculate
(a) NPV assuming a discount rate of 15%.
(b) Discounted cash flow rate of return (DCFROR)

(6 marks)

3. Assume that the copper concentrate contains 30% copper and 30 tr oz of silver per ton (consider ton here as st). It also contains 2% lead. All other deleterious elements are below the allowable limits. Assumed prices are $1/lb for copper, $6/tr oz for silver and $0.50/lb for lead.

Payments:
Copper: Percentage of contained metal (C)= 98%; price factor (f)= 1.0 and fixed unit deduction (u) = 1%
Silver: Percentage of contained metal (C)= 95%; price factor (f)= 1.0 and fixed unit deduction (u) = 1.0 tr oz
Deductions:
Copper: Treatment charge (T)= $75/ton; refining and selling cost (r)= $0.10/lb
Silver: Refining and selling cost (r)= $0.35 /tr oz of accountable silver
Assessments:
Lead: 1 unit is free and additional units charged at $10.00 per unit
Assume the concentrate shipped a distance (D) of 500 miles by rail to the smelter.
Transportation cost ($) per ton of concentrate is $F = 0.39 * D^{0.7}$
Calculate following values for per ton of concentrate
(i) Basic Smelter Return (BSR) (ii) Net Smelter Return (NSR) (iii) At-Mine Revenue (AMR) (iv) Gross Value of metal contained (v) Percent Payment (PP)

OR

Assume the following values to estimate (a) width of safety berm, (b) working bench width, (c) cut width, (d) bench face angle and (e) safety bench width. (f) Also, draw a section of bench indicating these. Assume other missing details wherever you find necessary.

i) Bench height = 40 ft.
ii) Angle of repose can be assumed to be 45° for berm materials
iii) The minimum clearance between the outer truck tire and the safety berm = 5 ft.
iv) Single spotting is used.
v) Bench face angle = 70°
vi) Loading is done with a 9 yd³ shovel with following specifications: dumping radius = 45.5 ft, maximum dumping height = 28 ft, level floor radius dimension = 25.25 ft, maximum shovel cutting radius = 54.5 ft.
vii) Haulage is by 85-ton capacity trucks.
viii) Truck width = 16 ft.
ix) Tyre rolling radius = 4 ft

(6 marks)

4. A mining operation has an annual sales revenue of Rs. 1,500,000 from a silver ore. Operating costs are Rs. 700,000, the allowable depreciation is Rs. 100,000 and the applicable tax rate is 32%. The cost depletion basis is zero. Calculate the cash flow with and without depletion [for depletion rate: choose smaller of (i) @50% of taxable income before depletion (ii) 15% of the gross revenue].

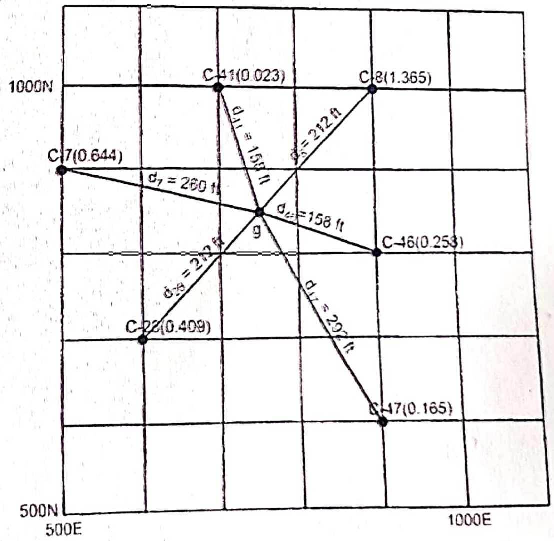
(4 marks)

5. **(i)** Draw a sequence of three benches. Label the crest, toe, bench face angle, bench width and bank width.

(ii) Assume that a 50 ft high bench has layering which goes through the toe. Assume that the following apply: $\phi = 32^{\circ}$, c = 100 kPa, $\rho = 2.45$ g/cm³, Bench face angle = 60°, Bedding angle = 20°. What is the safety factor?

(iii) A waste dump 500 ft high is planned. It is expected that the face angle will be 50°, the density is assumed to be 1.4 g/cm³, the cohesion = 0 and the friction angle is 30°. Will it be stable? Explain.

(iv) Using inverse distance squared (IDS) weighting formula, calculate grade at point 'g'.



(1.5 + 1.5 + 1.5 + 1.5 = 6 marks)

6. Discuss the problem associated with Floating Cone Technique when solving the two-dimensional Economic Block Model (refer Figure given below) assuming a 45° pit slope for estimation of ultimate pit limit.

	1	2	3	4	5
1	-1	-1	-4	-1	-1
2		+5	-4	+5	
3			+3		

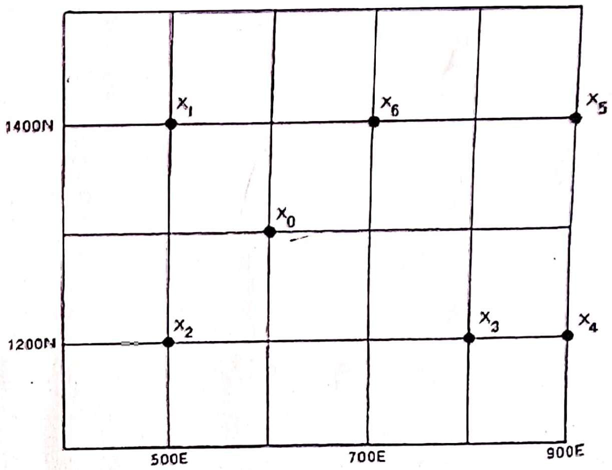
(3 marks)

7. The values of grade at $x_1 = 0.320\ (g_1)$ and $x_2 = 0.230\ (g_2)$. A spherical variogram has the following values: $c_o = 0.02;\ c_o + c_1 = 0.18; a = 450\ ft.$ Estimate the grade at x_0 using Kriging technique.

$$\gamma = \begin{cases} 0 & if\ h = 0 \\ c_1\left(\frac{3h}{2a} - \frac{h^3}{2a^3}\right) + c_o & if\ 0 < h \leq a \\ c_1 + c_o & if\ h > a \end{cases}$$

$$\sigma = \begin{cases} c_1 + c_o & if\ h = 0 \\ c_1 + c_o - \gamma & if\ 0 < h \leq a \\ 0 & if\ h > a \end{cases}$$

$$\sum_{j=1}^{n} a_j \sigma_{x_i x_j} + \lambda = \sigma_{x_0 x_i}$$



(4 marks)

Section B: Descriptive Answer Type Question (Answer in 5-6 lines/pointwise) – 28 marks

Answer all the questions and marks are assigned against each question

8. Differentiate between:
   i) depreciation and depletion
   ii) future worth and present value
   iii) payback period and rate of return
   iv) discounted cash flow and discounted cash flow rate of return
   v) metric ton, short ton and long ton
   vi) capital, operating and general administrative costs
   vii) net smelter returns and at-mine-revenue
   viii) safety bench and berms
   ix) single and double spotting of trucks
   x) switchback and ramp
   xi) drilling technique applicable for mineral exploration versus rock blasting
   xii) profits and revenues
   xiii) resources and reserves
   xiv) exploration, development and production
   xv) compositing and tonnage factor calculation

(15 marks)

9. Mention the different steps involved in estimation of ultimate pit limit using Lerchs-Grossmann 2-D Algorithm.

(2 marks)

10. Mine maps are the fundamental documents in all stages of mine planning and design. Discuss.

(2 marks)

11. Discuss the three stages of planning phase.

(1.5 marks)

12. Provide a checklist of factors to be addressed in the initial planning stages for any new mining project.

(1.5 marks)

13. Mining operations at any given mineral deposit may be divided into four stages. Discuss.

(1.5 marks)

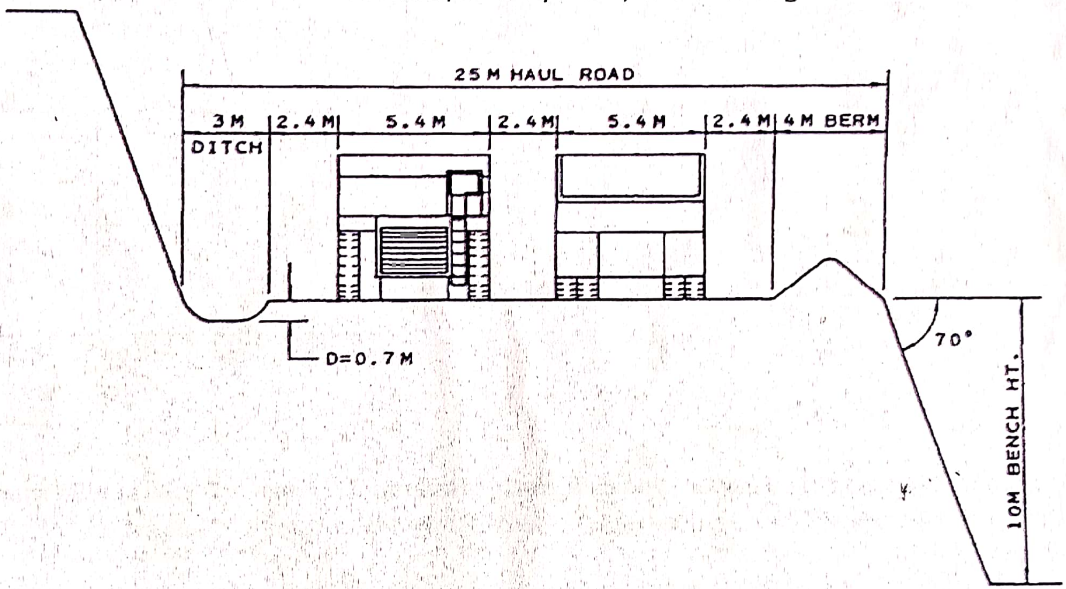
14. Draw curve of the different variogram models with their applicability conditions.

(1.5 marks)

15. Discuss some of the aspects that enter into bench height selection.

(1.5 marks)

16. Discuss the importance of travel lane width, a safety berm, and a drainage ditch.



(1.5 marks)

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